

EULERIAN AND SEMI-EULERIAN GRAPHS

Eulerian Trail

An Eulerian trail is a trail in a connected graph that travels every edge in the graph only once. Note that repeated vertices are permitted but not repeated edges.

Eulerian Circuit

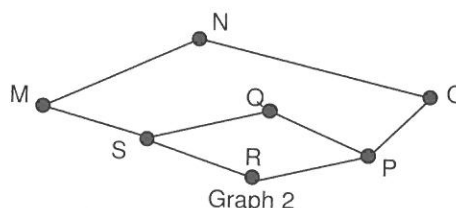
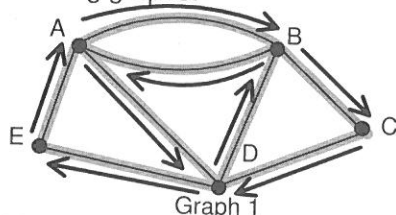
If an Eulerian trail is **closed**, that is it begins and ends at the same vertex, it is said to be an Eulerian Circuit.

Eulerian Graph

A connected graph is said to be Eulerian graph if it has a closed trail (starts and ends at the same vertex) and includes every edge once only. Or simply a graph is considered to be **Eulerian** if you can start at a vertex, traverse through every edge only once, and return to the same vertex you started at.

Eulerian graphs have **only even** vertices, that is, no odd vertices.

Consider the following graphs:



Graph 1 is an Eulerian graph because you can start at any vertex traverse each edge only once and return to the starting vertex. Note all vertices are even.

The trail ABCDEADBA is a closed trail, it starts and ends at the same vertex and traverses through every edge once only and hence it is an Eulerian circuit. The Eulerian circuit ABCDEABA is not unique.

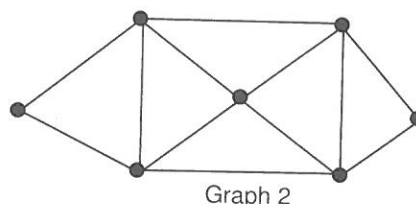
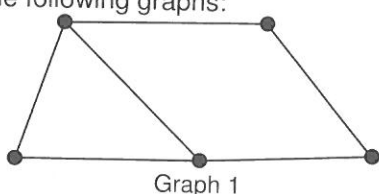
Graph 2 is not an Eulerian graph as there is no Eulerian circuit. You cannot start at a vertex, traverse each edge only once and return to the starting vertex. Note all vertices are not even.

Determining if a Graph is Eulerian

For a connected graph to be Eulerian and an Eulerian circuit to exist, the following conditions must apply:

- every vertex must be of even degree
- you must start and finish at the same vertex
- every edge must be traversed only once

Consider the following graphs:



Graph 1 is not Eulerian as it has two vertices with odd degree.

Graph 2 is Eulerian as each vertex has an even degree.

Semi-Eulerian Trail

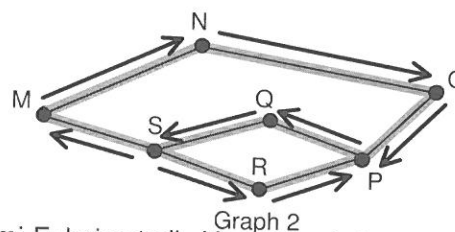
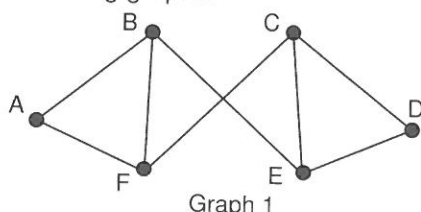
If an Eulerian trail is **open**, that is, it begins and ends at different vertices, it is said to be a semi-Eulerian trail.

Semi-Eulerian Graphs

A connected graph is said to be semi-Eulerian graph if it has a semi-Eulerian trail. Or simply a graph is considered to be semi-Eulerian if you can start at a vertex, traverse through every edge only once, and end up at a different vertex.

Semi-Eulerian graphs must have exactly **two odd vertices**.

Consider the following graphs:



Graph 1 is not a semi-Eulerian graph as there is no semi-Eulerian trail. You cannot start at a vertex, traverse each edge only once and end at a different vertex. Note that graph 1 has more than two odd vertices.

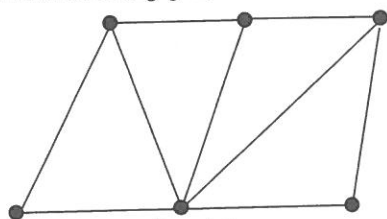
Graph 2 is a semi-Eulerian graph. If you start at one of the odd degree vertices all edges can be traversed once only and you finish at the other odd degree vertex completing a semi-Eulerian trail. For example SMNOPQSRP is a semi-Eulerian trail, hence graph 2 is a semi-Eulerian graph. Note that graph 2 has exactly two odd vertices.

Determining if a Graph is semi-Eulerian

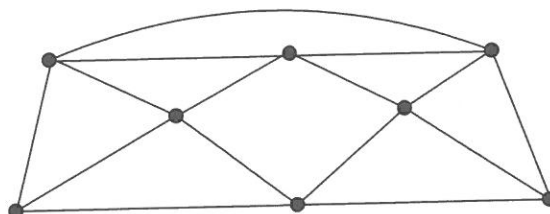
For a connected graph to be semi-Eulerian and a semi-Eulerian trail to exist, the following conditions must apply:

- only two vertices can be of odd degree, every other vertex must be of even degree
- you must start the semi-Eulerian trail at a vertex with odd degree and end at the other vertex with odd degree
- every edge must be traversed only once

Consider the following graphs:



Graph 1



Graph 2

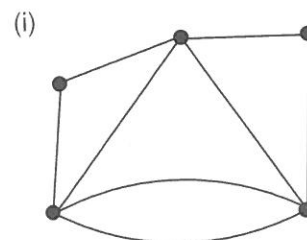
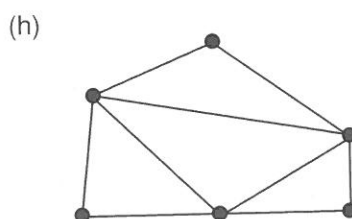
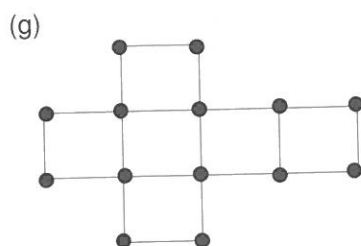
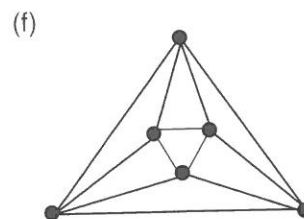
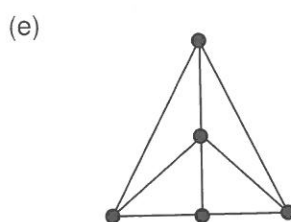
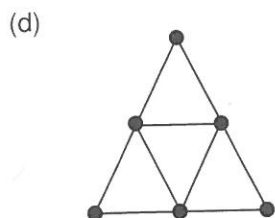
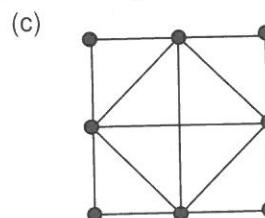
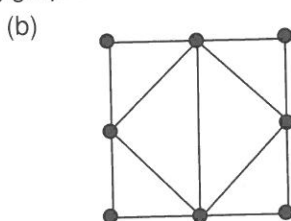
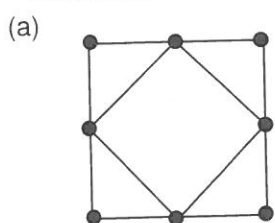
Graph 1 is not Eulerian as it has vertices with odd degree, and it is not semi-Eulerian as it has more than two vertices with odd degree.

Graph 2 is not Eulerian as it has vertices with odd degree, however it is semi-Eulerian as it has exactly two vertices with odd degree and all other vertices are of even degree.

- Note:
- For a graph or network to be traversable it must be connected and have either zero odd vertices or exactly two odd vertices.
 - For a Eulerian trail (closed trail) and hence a Eulerian graph each edge must be traversed only once, vertices may be traversed more than once and the connected graph must have zero odd vertices.
 - For a Semi-Eulerian trial (open trail) and hence a semi-Eulerian graph each edge must be traversed only once, vertices may be traversed more than once and the connected graph must have exactly two odd vertices. The semi-Eulerian trail will start at one of the odd vertices and finish at the other odd vertex.

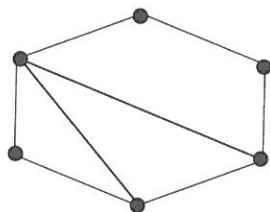
EXERCISE 8B

1. Determine which of the following graphs are Eulerian. If not Eulerian state why.

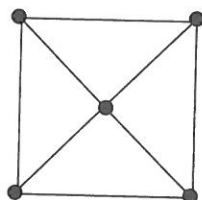


2. Determine which of the following graphs are Eulerian, Semi-Eulerian or neither. If the graph is neither Eulerian or Semi-Eulerian state why.

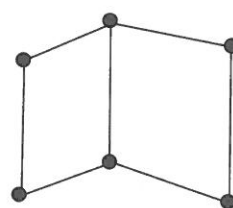
(a)



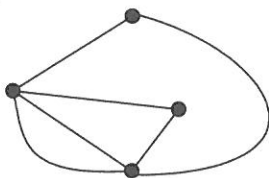
(b)



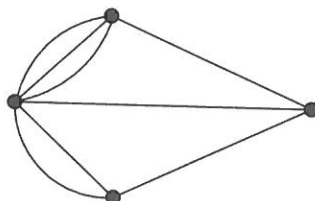
(c)



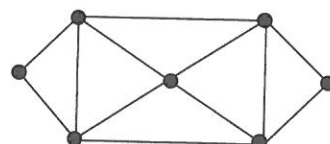
(d)



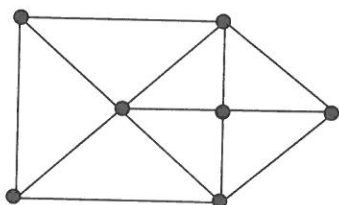
(e)



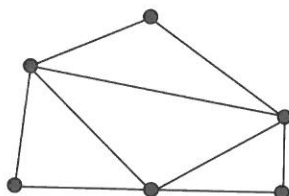
(f)



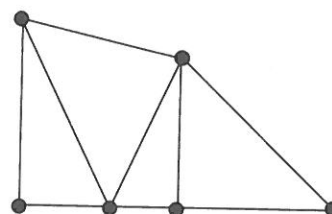
(g)



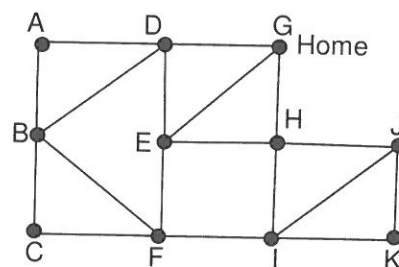
(h)



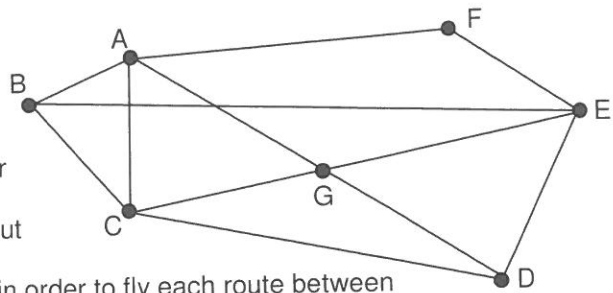
(i)



3. Which of the graphs in question 2 can be traversed using each edge only once?
4. Consider the given information and decide whether the graph is Eulerian, semi-Eulerian or neither.
- (a) The graph has only even vertices. (b) A trail of the graph starts and finished at different vertices.
- (c) The graph has no vertices of odd degree. (d) An open walk passes through all edges only once.
- (e) The graph has more even vertices than odd vertices. (f) The graph has five even vertices and only one odd vertex.
- (g) A walk passes through all vertices exactly once. (h) A walk passes through all edges exactly once.
- (i) A closed walk passes through all edges. (j) The graph has only two odd vertices.
5. The graph represents the local area street map for the delivery of the community newspaper with vertices representing street intersections and edges the roads between them. Joshua's home is located at G and he needs to plan a route so that he does not travel the same road twice.
- (a) Can Joshua start and finish the delivery of the newspaper at his home? Justify your answer.
- (b) If Joshua starts the delivery round at his home where will he finish?
- (c) A new road is built joining G and J. Is it possible for Joshua to start and finish the delivery at his home? Justify your answer.
- (d) State a suitable delivery route for Joshua.



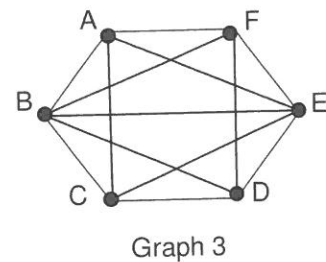
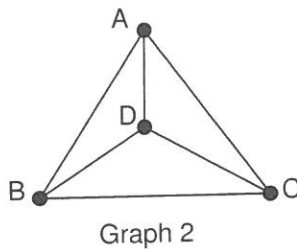
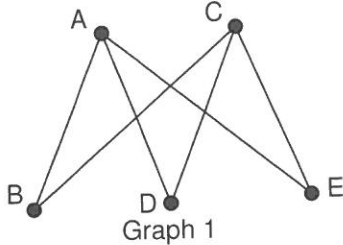
6. The maps that pilots use are called route maps. Route maps show the flight paths of aeroplanes from one destination to another. The route map on the right shows the flight paths between cities labelled A to G for a particular airline company.



A trainee pilot needs to fly all of these routes without flying the same route more than once.

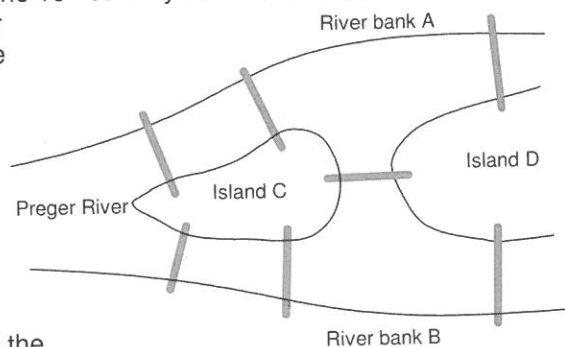
- (a) Where should the trainee pilot start and finish in order to fly each route between these cities without flying the same route twice? Justify mathematically.
- (b) What minimum change needs to be made to the route map by the airline company to enable trainee pilots to fly each route and return to the city they started at? Justify mathematically.

7. Consider the following graphs.



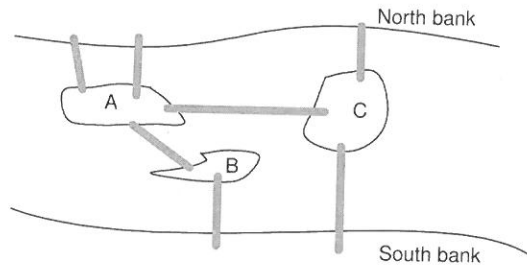
- (a) Which of the given graphs are semi-Eulerian? Justify your response.
- (b) For each semi-Eulerian graph write down a corresponding semi-Eulerian trail.

8. Königsberg is a town on the Preger River, which in the 18th century was a German town, but now is Russian. The river Preger divides the town into four sections, A, B, C and D as shown by the map on the right. There are seven bridges connecting these sections and many citizens wondered whether they can walk across all seven bridges without crossing any bridge more than once.



- (a) Represent the map of the townsites as a graph in the space below the map.
- (b) Is it possible to visit each of the four sections of the town by crossing every bridge once and once only? Justify mathematically.
- (c) If an extra bridge was built would solve the problem? Where should it be built? Justify your response.
- (d) If one of the existing bridges was removed would this solve the problem? Which bridge should be removed?

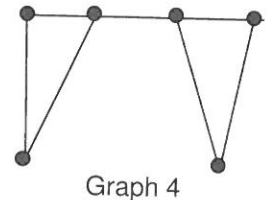
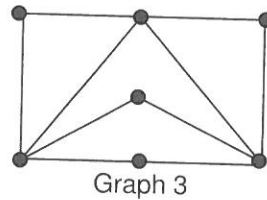
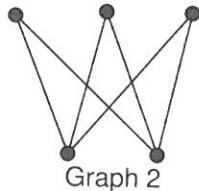
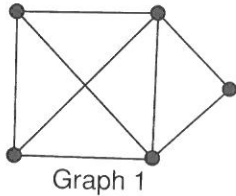
9. In a certain city there is a river running through the middle of the city. There are three islands and seven bridges connecting the island and the mainland as shown on the right.



- (a) Construct a graph to represent this information.
- (b) Is it possible to start at the North bank cross all of the bridges only once and finish at the North Bank? Justify your answer.

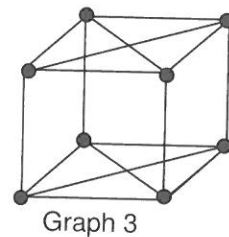
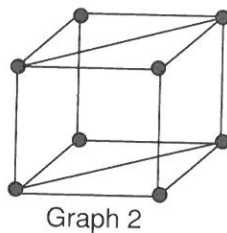
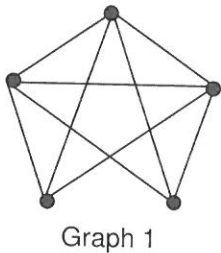
- (c) Is it possible to cross all bridges only once? If so, where should you start and where will you finish?

10. Consider the following graphs.



- (a) Which graphs are connected?
- (b) Which graphs are planar?
- (c) Which graphs are complete?
- (d) Which graph has an Euler Circuit?
- (e) Which graphs have a semi-Eulerian trail?
- (f) Which graph has an Eulerian trail?
- (g) Which graph has a bridge?
- (h) Which graphs do not have a cycle?

11. For each of the following four graphs decide if it has an Eulerian trail, semi-Eulerian trail, Eulerian circuit or neither. You must **justify** your choice.



Graph of K_{10}

Graph 4